

1.3, 2.5

4.1 Answer each of the following:

- a. Suppose that a simple regression has quantities $N = 20$, $\sum y_i^2 = 7825.94$, $\bar{y} = 19.21$, and $SSR = 375.47$, find R^2 .
- b. Suppose that a simple regression has quantities $R^2 = 0.7911$, $SST = 725.94$, and $N = 20$, find $\hat{\sigma}^2$.
- c. Suppose that a simple regression has quantities $\sum (y_i - \bar{y})^2 = 631.63$ and $\sum e_i^2 = 182.85$, find R^2 .

a.

$$SS_T = 7825.94 - 20 \times 19.21^2 = 445.458$$

$$R^2 = \frac{SSR}{SS_T} = \frac{375.47}{445.458} = 0.8429$$

b.

$$\hat{\sigma}^2 = \frac{SSR}{N-2}$$

$$SSR = 445 - 375.47$$

$$R^2 = \frac{SSR}{SS_T} \Rightarrow 0.7911 = \frac{SSR}{725.94} \Rightarrow SSR = 574.2911$$

$$SSR = 725.94 - 574.2911 = 151.6489$$

$$\hat{\sigma}^2 = \frac{151.6489}{18} = 8.4249$$

c.

$$R^2 = \frac{SS_T - SSR}{SS_T} = \frac{631.63 - 182.85}{631.63} = 0.7105$$

4.3 We have five observations on x and y . They are $x_i = 3, 2, 1, -1, 0$ with corresponding y values $y_i = 4, 2, 3, 1, 0$. The fitted least squares line is $\hat{y}_i = 1.2 + 0.8x_i$, the sum of squared least squares residuals is $\sum_{i=1}^5 e_i^2 = 3.6$, $\sum_{i=1}^5 (x_i - \bar{x})^2 = 10$, and $\sum_{i=1}^5 (y_i - \bar{y})^2 = 10$. Carry out this exercise with a hand calculator. Compute

- a. the predicted value of y for $x_0 = 4$.
- b. the $se(f)$ corresponding to part (a).
- c. a 95% prediction interval for y given $x_0 = 4$.
- d. a 99% prediction interval for y given $x_0 = 4$.
- e. a 95% prediction interval for y given $x = \bar{x}$. Compare the width of this interval to the one computed in part (c).

a.

$$\hat{y}_0 = 1.2 + 0.8 \times 4 = 4.4$$

b.

$$se(f) = \sqrt{\hat{\sigma}^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum x_i^2} \right)}$$

$$\hat{\sigma}^2 = \frac{3.6}{5} = 0.72, \bar{x} = \frac{3+2+1+0+(-1)}{5} = 1$$

$$se(f) = \sqrt{0.72 \left(1 + \frac{1}{5} + \frac{(4-1)^2}{10} \right)} = \sqrt{0.72 \times \frac{21}{10}} = 1.5875$$

c.

$$(1.2 + 0.8 \times 4) \pm t_{0.025}(3) \cdot 1.5875, t_{0.025}(3) = 3.182 \Rightarrow [4.4 - 1.5875 \times 3.182, 4.4 + 1.5875 \times 3.182] = [-0.6514, 9.4514]$$

d.

$$(1.2 + 0.8 \times 4) \pm t_{0.005}(3) \cdot 1.5875, t_{0.005}(3) = 5.841 \Rightarrow [4.4 - 1.5875 \times 5.841, 4.4 + 1.5875 \times 5.841] = [-4.8926, 13.6726]$$

e.

$$(1.2 + 0.8 \times 1) \pm t_{0.025}(3) \sqrt{0.72 \left(1 + \frac{1}{5} \right)} \Rightarrow [2 - 3.182 \cdot \sqrt{1.2 \times \frac{6}{5}}, 2 + 3.182 \cdot \sqrt{1.2 \times \frac{6}{5}}] = [-1.819, 5.819], \text{ (e) is shorter}$$